FROM A CANDIDATE TO A PHYSICAL CLAIM

Biswajit Jana | JWST Cosmic Anomaly Lab | Research progress: 12 November 2026

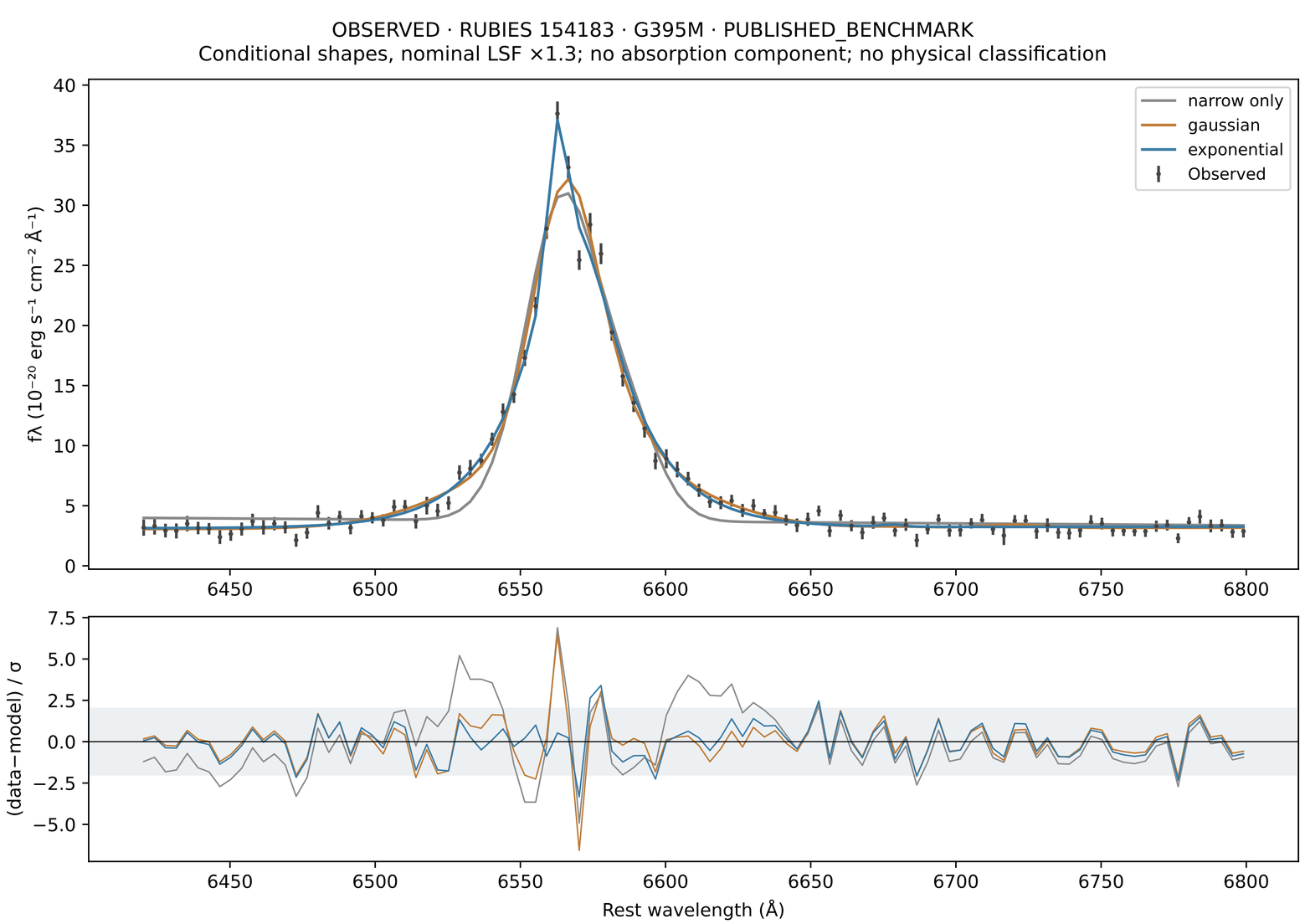
CONFERENCE POSTER 02

CEERS / JADES / RUBIES / UNCOVER

Public observations. Reproducible analysis.

No new black-hole-star identification.

THE CLIFF / RUBIES-UDS-154183



A missed-benchmark check: the PRISM continuum screen finds The Cliff, but the fixed-width broad-blend screen misses it. The 11-source queue is therefore incomplete for published candidates. 15 sources received detailed inspection; 7 yielded conditional medium-resolution fits. No new source is established as a black-hole star.

01 / WHAT THE DATA PREFER

The Cliff is a **published candidate**, added separately as a benchmark. Our G395M fit compares a Gaussian broad component with a Gaussian convolved with symmetric exponential tails. Narrow nuisance lines and a local continuum are included.

67.6

Full-window delta BIC

9.0

With four core pixels masked

02 / THE SENSITIVITY TEST

Positive delta BIC favours exponential tails. Across LSF tests, the full-window value is 56.8-69.6, but it falls to 3.5-9.4 after excluding the central +/-400 km/s. With doubled errors, the masked comparison no longer favours the exponential family.

03 / WHAT THIS DOES NOT SHOW

The preference depends on the line core, where absorption is unmodelled. Empirical shapes cannot uniquely identify an accretion envelope. No black-hole mass, density or physical-class probability is inferred here.

04 / THE NEXT PHYSICAL TEST

Fit absorption and emission together; validate extraction and calibration; compare stellar, AGN and dense-envelope predictions across multiple independent observables.

Benchmark: de Graaff et al. (2025), A&A; 701, A168, arXiv:2503.16600. Observations: RUBIES / JWST; reduction: DJA / msaexp. This project performs a conditional reanalysis, not an independent reduction. Results and executable methods: github.com/Biswajit1999/JWST-Cosmic-Anomaly-Lab